Astrodynamics II

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Chapter 6. Orbital Maneuvering

6.0 Preliminary

- Discuss the motivation of orbital maneuvering
  - Orbit Change: Coplanar and Non-coplanar
  - Orbit Keeping: Perturbation
  - Interplanetary/Interlunar Maneuvers

- Discuss orbital maneuvers by impulsive/continuous thrust
  - Impulsive Thrust
    - (High thrust) burns are assumed to be instantaneous
    - No changes in positions, instantaneous change in velocities
    - In reality, impulsive burns are to be corrected to include finite duration effects
    - May use unforced dynamic equations of motion
  - Continuous Thrust
    - Burns are assumed to last for finite time
    - Usually use the dynamics which contain the control (control input, force, actuation) term usually denoted by $\boldsymbol{u}$

- Optimality
  - Usually means fuel, time, and their combinations
  - When/where/which direction/how big the burns are for the required maneuver
  - Rigorous studies are beyond the scope of this course
  - Will be studied in Graduate Course

6.1 Historical Background

- **Orbital Maneuvers** encompass all orbital changes after insertion required to place a satellite in the orbit we choose.

- The early history of **maneuvers** is sometimes centered on the first experiments with rockets:
  - Konstantin Tisolkovsky (1857-1935)
  - Herman Oberth (1819-1989)
  - Robert Goddard (1882-1945)
  - Wernher von Braun (1912-1977)

- Actual **satellite maneuvering** occurred after the launch of Sputnik on October 4, 1957. Thus, orbital maneuvering had its roots in the classical formulas from several centuries ago, but didn't mature until recently.
  - Luna 1 (Soviet, Jan. 2, 1959)
    - Mission to the Moon (missed the Moon by 6,000 km)
    - The first satellite maneuvers
  - Luna 2 (Soviet, Sept. 12, 1959)
    - Crashed into lunar surface
    - No retro-rocket technology for a soft landing

- After some initial failures at landing on the Moon, the focus of scientists shifted to near-Earth operations, for weather forecasting, communications, etc. **Maneuvers** were used to place each of these satellites (designed for a specific mission and orbit) into the correct orbit.
  - Voskhod 2 (March 18, 1965): the first human walking in space
  - Gemini 3 (March 23, 1965): the first actual maneuver for a manned S/C
  - Gemini 6 and 7 (Dec. 1965): the first rendezvous
  - Apollo and Soyuz (July 1975), Space Shuttle and Mir (1995): rendezvous mission with two spacecraft from different countries

6.2 Introduction

- Orbital Maneuvers with instantaneous burns change various Orbital Elements
  - A force applied in the orbital plane changes *e, a, w*
  - A force applied normal to the orbital plane changes *i, Ω*

- A goal of any maneuver, particularly those in which vehicle weight is critical, is efficiency
  - Efficiency can mean using a minimum amount of fuel, while satisfying the mission requirements usually including operational time span
  - The change in velocity (Δv) enables one to estimate the fuel required for orbital maneuvers by using the **Rocket Equation**:

$$\Delta v = V_e \, ln\left(\frac{m_0}{m_f}\right), \quad Rocket\ eq.$$

$$\frac{\Delta v}{V_e} = -ln\left(\frac{m_f}{m_0}\right), \qquad e^{-\frac{\Delta v}{V_e}} = \frac{m_f}{m_0}, \quad m_f = m_0 e^{-\frac{\Delta v}{V_e}}$$

$$\Delta m = m_0 - m_f = m_0\left(1 - e^{-\frac{\Delta v}{V_e}}\right), \qquad where \quad V_e = I_{sp} g$$

- The vis-viva equation allows us to calculate Δv:

$$E = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} \qquad \rightarrow \qquad v^2 = \mu(\frac{2}{r} - \frac{1}{a}) \quad vis-viva \;\; equation$$

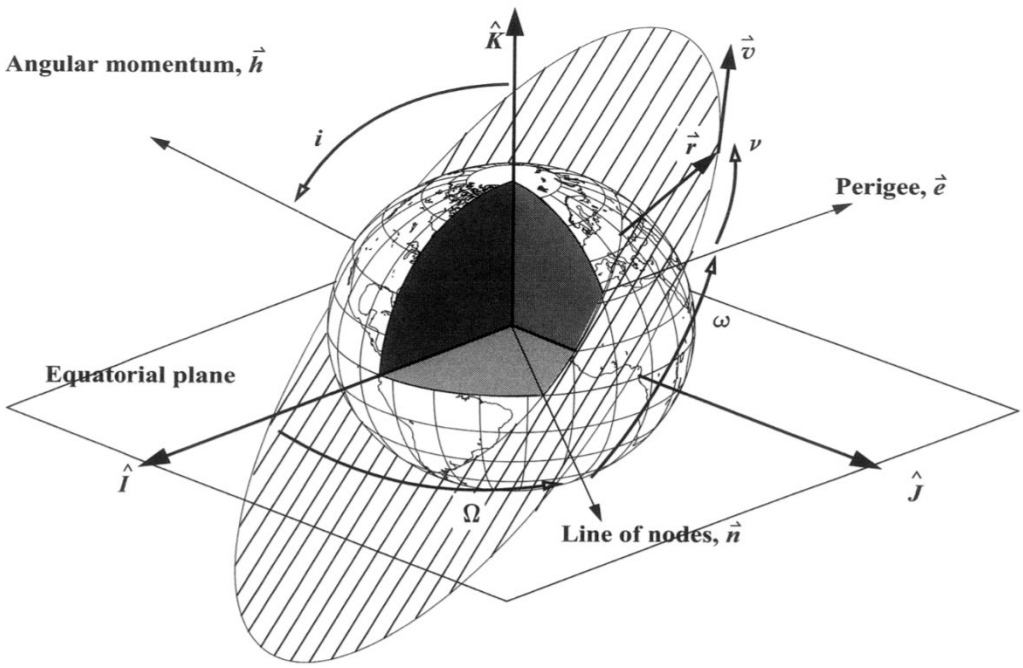


Figure 2-16. Classical Orbital Elements. The six classical orbital elements are the *semimajor axis, a*; *eccentricity, e*; *inclination, i*; *right ascension of the ascending node*, Ω—often referred to as simply the *node*; *argument of perigee*, ω; and *true anomaly*, ν. I haven't shown scale and shape elements (*a* and *e*) because I've introduced them in Chap. 1.

6.3 Coplanar Maneuvers

- **Coplanar Maneuvers** do not change the orbital plane:
  - Initial & Final orbits lie in the same plane
  - Change the orbit's size & shape (a, e) and the location of the line of apsides (ω)
  - Either tangential or non-tangential
    - For maximum efficiency of the burn, the flight-path angle must be zero (tangential burn)
    - For elliptical orbits, the tangential burns with zero flight-path angle occur only at apoapsis & periapsis.

- Three types of Coplanar Maneuvers
  - Hohmann Transfer (two tangential burns)
  - One Tangential Burn, one non-tangential burn
  - General Transfers (two non-tangential burns)

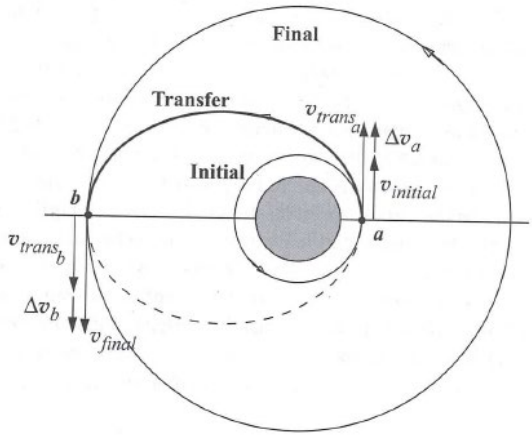


Figure 6-1. **Orbit Notation.** The letters a, b, … represent changes in velocities and locations of the firings. When we use multiple transfer orbits, numbers signify which transfer we're considering.

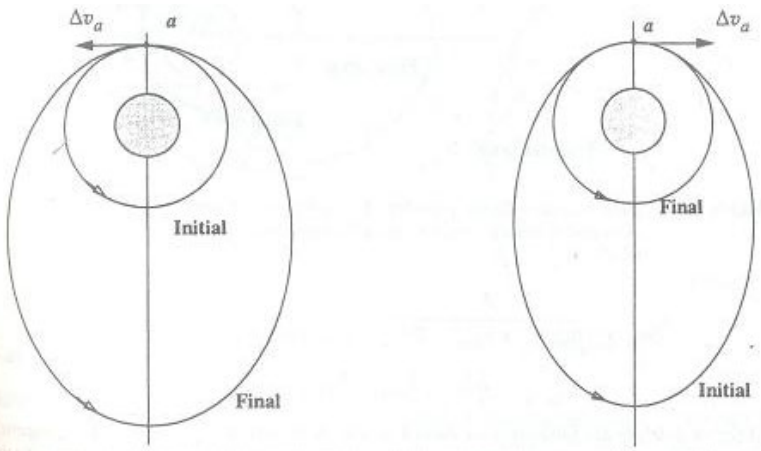


Figure 6-2. **Tangential Orbit Transfer ($\phi_{fpa} = 0°$).** Tangential burns are defined to occur with a zero flight-path angle. Velocity increases enlarge the semimajor axis (left), whereas decreases cause the orbit to come closer to the Earth (right).

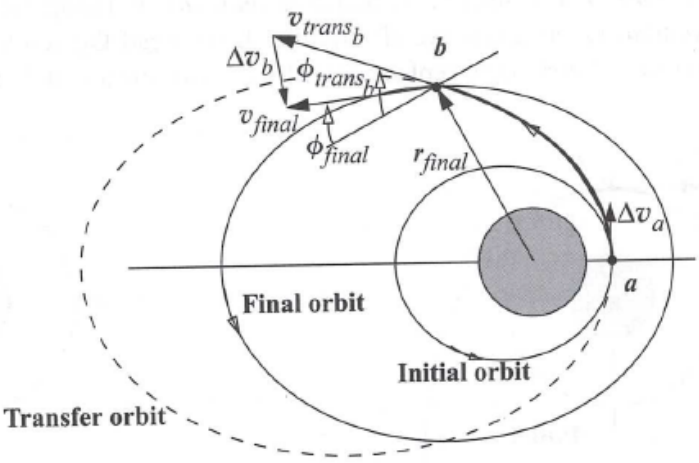


Figure 6-3. **Nontangential Orbit Transfer ($\phi_{fpa} \neq 0$).** The nontangential transfer, point b, has nonparallel velocity vectors because both orbits have flight-path angles greater than zero.

6.3.1 Hohmann Transfer and Bi-elliptic Transfer

- Hohmann Transfer
  - The minimum energy transfer can be achieved between orbits by using two tangential burns:
    - Given: $r_{initial}$, r_{final}, $a_{initial}$, a_{final}
    - Calculate: a_{trans}, τ_{trans}, Δv_a, Δv_b

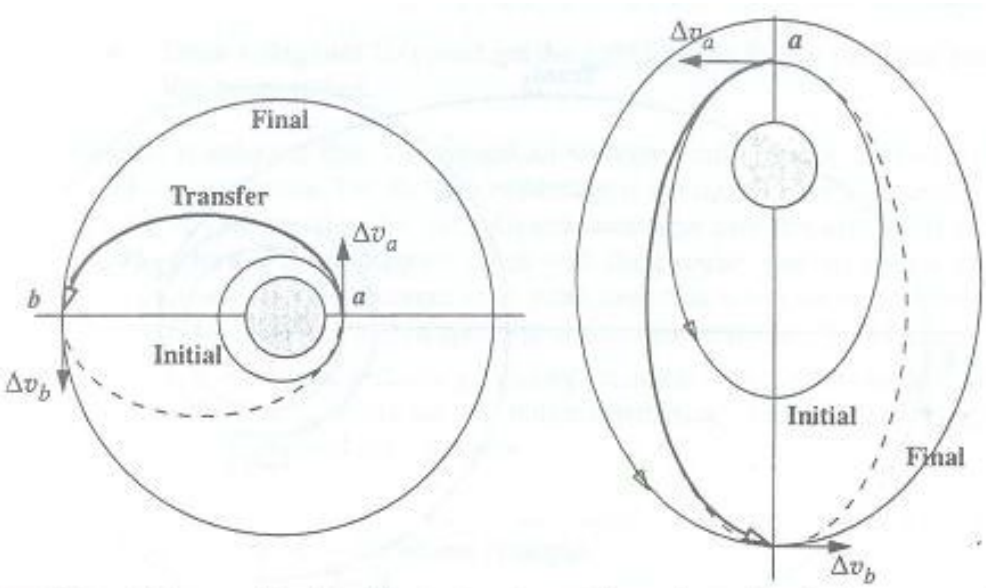


Figure 6-4. **Hohmann Transfer.** The two-impulse minimum change in velocity maneuver is the Hohmann transfer. Elliptical transfers must start at either apogee or perigee for this maneuver to be a minimum.

$$a_{trans} = \frac{r_{initial} + r_{final}}{2}$$

$$v_{initial} = \sqrt{\frac{2\mu}{r_{initial}} - \frac{\mu}{a_{initial}}} \qquad v_{trans,a} = \sqrt{\frac{2\mu}{r_{initial}} - \frac{\mu}{a_{trans}}}$$

$$v_{trans,b} = \sqrt{\frac{2\mu}{r_{final}} - \frac{\mu}{a_{trans}}} \qquad v_{final} = \sqrt{\frac{2\mu}{r_{final}} - \frac{\mu}{a_{final}}}$$

$$\Delta v_a = v_{trans,a} - v_{initial} \qquad \Delta v_b = v_{final} - v_{trans,b}$$

$$\Delta v = |\Delta v_a| + |\Delta v_b|$$

$$\tau_{trans} = \frac{T_{trans}}{2} = \pi\sqrt{\frac{a_{trans}^3}{\mu}} \qquad where\ T_{trans} = 2\pi\sqrt{\frac{a_{trans}^3}{\mu}}$$

- Bi-elliptic Transfer
  - The bi-elliptic transfer employs two Hohmann transfers in succession
  - Using three burns adds complexity, but can save Δv in some cases
    - Given: $r_{initial}$, r_b, r_{final}
    - Calculate: a_{trans1}, τ_{trans1}, a_{trans2}, τ_{trans2}, Δv_a, Δv_b, Δv_c

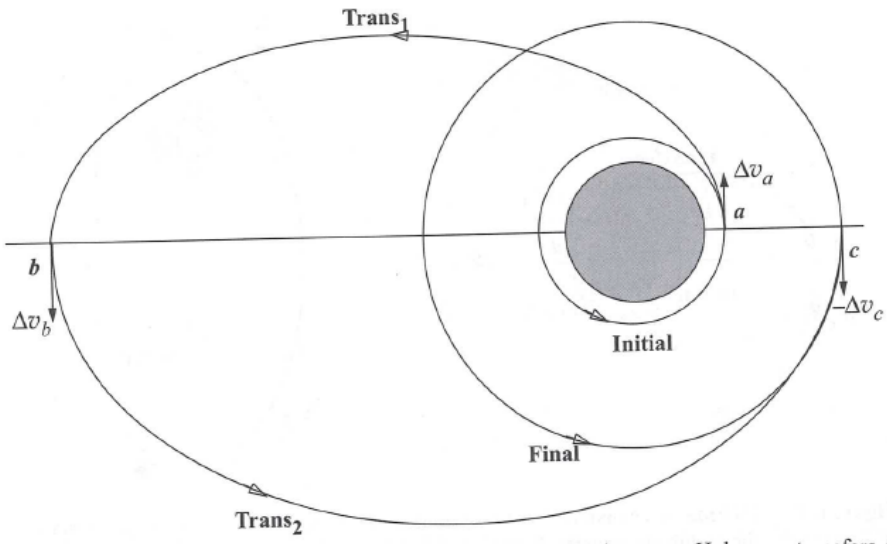


Figure 6-5. **Bi-elliptic Transfer.** The bi-elliptic transfer employs two Hohmann transfers in succession. This allows us to combine the middle two burns. Using three burns adds complexity but can save Δv in some instances.

$$a_{trans1} = \frac{r_{initial} + r_b}{2} \qquad a_{trans2} = \frac{r_b + r_{final}}{2}$$

$$v_{initial} = \sqrt{\frac{\mu}{r_{initial}}} \qquad (circular\ orbit)$$

$$v_{trans1,a} = \sqrt{\frac{2\mu}{r_{initial}} - \frac{\mu}{a_{trans,1}}} \qquad v_{trans1,b} = \sqrt{\frac{2\mu}{r_b} - \frac{\mu}{a_{trans,1}}}$$

$$v_{trans2,b} = \sqrt{\frac{2\mu}{r_b} - \frac{\mu}{a_{trans,2}}} \qquad v_{trans2,c} = \sqrt{\frac{2\mu}{r_{final}} - \frac{\mu}{a_{trans,2}}}$$

$$v_{final} = \sqrt{\frac{\mu}{r_{final}}} \quad (circular\ orbit)$$

$$\Delta v_a = v_{trans1,a} - v_{initial} \;;\; \Delta v_b = v_{trans2,b} - v_{trans1,b} \;;\; \Delta v_c = v_{trans2,c} - v_{final}$$

$$\Delta v = |\Delta v_a| + |\Delta v_b| + |\Delta v_c|$$

$$\tau_{trans,1} = \frac{T_{trans,1}}{2} = \pi\sqrt{\frac{a_{trans,1}^3}{\mu}} \qquad \tau_{trans,2} = \frac{T_{trans,2}}{2} = \pi\sqrt{\frac{a_{trans,2}^3}{\mu}}$$

$$\tau = \tau_{trans,1} + \tau_{trans,2}$$

6.3.2 Comparing Hohmann and Bi-elliptic Transfers

- The Hohmann transfer is the minimum Energy transfer between most, but not all, coplanar orbits. In some cases, the bi-elliptic transfer may use less energy.

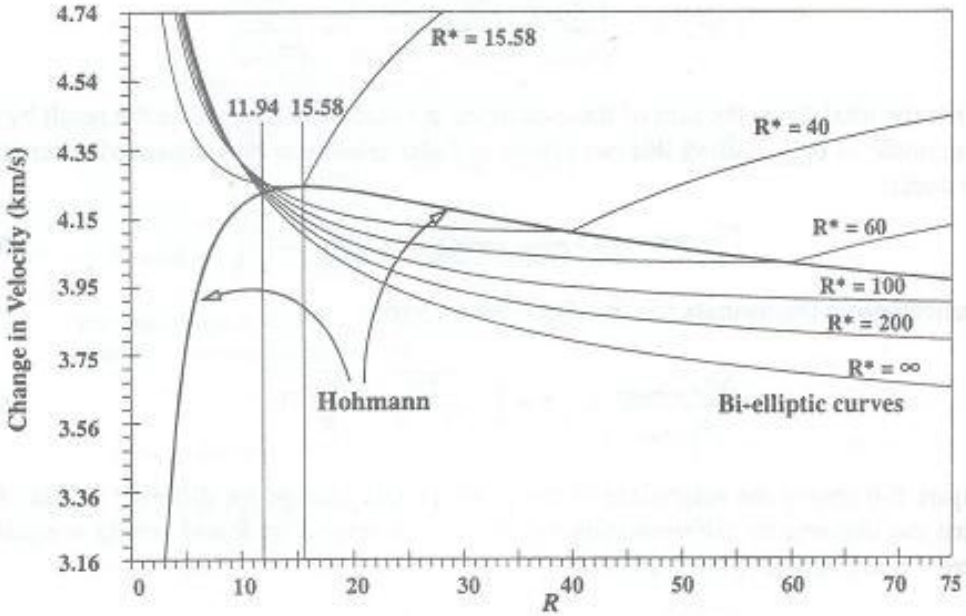


Figure 6-6. **Hohmann Transfer—Comparison of Change in Velocity.** The change in velocity is shown when using Hohmann and bi-elliptic orbit changes. Be aware of the different notations for the ratios (final / initial) (R for Hohmann and R^* for bi-elliptic).

$$R = \frac{r_{final}}{r_{initial}} \left(= \frac{a_{final}}{a_{initial}} \right) \quad \textit{for circular orbits}$$

$$R^* = \frac{r_b}{r_{initial}}$$

I. **0 < R < 11.94** ; the Hohmann transfer is superior!

II. **11.94 < R < 15.58** ; we need another test to see which method is best for realistic transfer.

III. **15.58 < R <** ∞ ; the Bi-elliptic method can be superior, although we may need a very large intermediate orbit, significantly increasing the transfer time.

IV. **If the final radius is larger than both the initial and transfer orbits**, the Hohmann transfer is always superior in terms of total change in velocity.

- For Bi-elliptic transfer, at $r_b=r_{final}$, direction of the second burn changes, and thus non-smooth connection at the point
- For Hohmann transfer, a larger R does not necessarily imply a larger Δv

6.3.3 Transfers using the One-Tangent Burn (& one non-tangential burn)

- One-Tangent Burn reduces the transfer time of the Hohmann techniques but increases Δv requirements
- Δv burn is performed when two orbits are intersected.
- A one-tangent burn maneuver consists of one tangential burn and one non-tangential burn

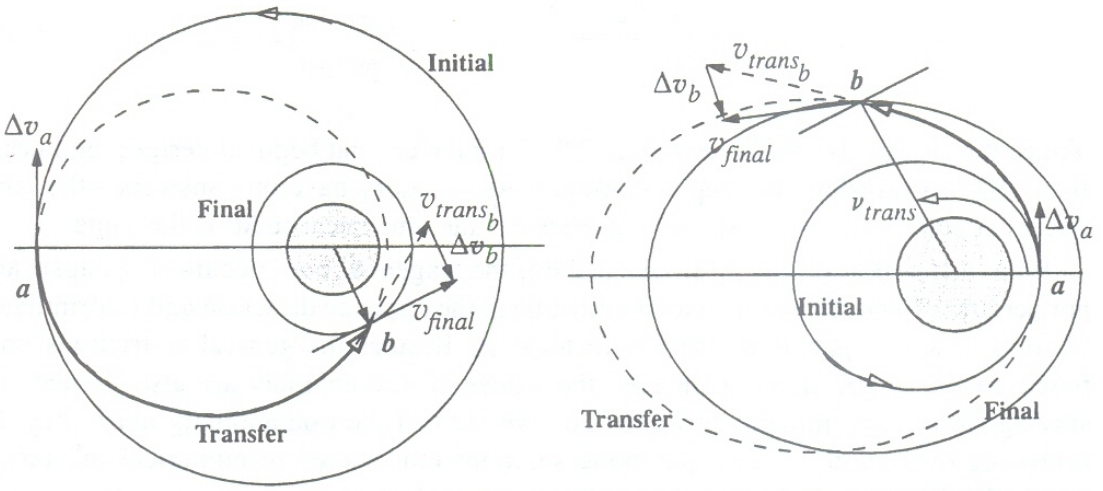


Figure 6-7. **One-Tangent Burns.** For both circular (left) and elliptical (right) orbit changes, the true anomalies of the transfer and final orbits are equal. The maneuver starts at apoapsis (left, $\nu > 180°$) and periapsis (right, $\nu < 180°$) of the transfer orbit.

- Recall: Perigee Distance: $r_p=a(1-e)$, Apogee Distance: $r_a=a(1+e)$, $r=\dfrac{a(1-e^2)}{1+e\cos\nu}$
  - Tangential Burn at Periapsis (right figure of Fig 6-7)

$$r_{initial} = a_{trans}(1-e_{trans}) \Rightarrow a_{trans} = \frac{r_{initial}}{1-e_{trans}}$$

$$r_{final} = \frac{a_{trans}(1-e_{trans}^2)}{1+e_{trans}\cos\nu_{trans,b}} = \frac{r_{initial}(1+e_{trans})}{1+e_{trans}\cos\nu_{trans,b}}$$

$$\Rightarrow r_{final} + r_{final}e_{trans}\cos\nu_{trans,b} = r_{initial} + r_{initial}e_{trans}$$

$$\Rightarrow e_{trans}(r_{final}\cos\nu_{trans,b} - r_{initial}) = r_{initial} - r_{final}$$

$$\Rightarrow e_{trans} = \frac{R^{-1}-1}{\cos\nu_{trans,b} - R^{-1}},\quad R^{-1} = \frac{r_{initial}}{r_{final}}$$

- Tangential Burn at Apoapsis (left figure of Fig 6-7)

$$r_{initial} = a_{trans}(1+e_{trans}) \Rightarrow a_{trans} = \frac{r_{initial}}{1+e_{trans}}$$

$$r_{final} = \frac{a_{trans}(1-e_{trans}^2)}{1+e_{trans}\cos\nu_{trans,b}} = \frac{r_{initial}(1-e_{trans})}{1+e_{trans}\cos\nu_{trans,b}}$$

$$\Rightarrow r_{final} + r_{final}e_{trans}\cos\nu_{trans,b} = r_{initial} - r_{initial}e_{trans}$$

$$\Rightarrow e_{trans}(r_{final}\cos\nu_{trans,b} + r_{initial}) = r_{initial} - r_{final}$$

$$\Rightarrow e_{trans} = \frac{R^{-1}-1}{\cos\nu_{trans,b} + R^{-1}}, \quad R^{-1} = \frac{r_{initial}}{r_{final}}$$

- Circular Case
  - Given $r_{initial}$, r_{final}, $\nu_{trans,b}$
  - Calculate e_{trans}, a_{trans}, τ_{trans}, Δv_a, Δv_b

$$e_{trans} = \frac{R^{-1}-1}{\cos\nu_{trans,b} \mp R^{-1}} \quad \begin{Bmatrix} - & periapsis & for\,tangential & burn \\ + & apoapsis & for\,tangential & burn \end{Bmatrix}, \quad R^{-1} = \frac{r_{initial}}{r_{final}}$$

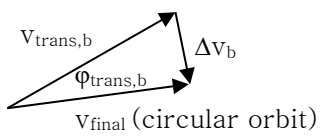
$$a_{trans} = \frac{r_{initial}}{1 \mp e_{trans}}$$

$$v_{initial} = \sqrt{\frac{\mu}{r_{initial}}}, \qquad v_{trans,a} = \sqrt{\frac{2\mu}{r_{initial}} - \frac{\mu}{a_{trans}}}$$

$$\Rightarrow \Delta v_a = v_{trans,a} - v_{initial}$$

$$v_{final} = \sqrt{\frac{\mu}{r_{final}}}, \qquad v_{trans,b} = \sqrt{\frac{2\mu}{r_{final}} - \frac{\mu}{a_{trans}}}$$

$$\tan\phi_{trans,b} = \frac{e_{trans}\sin\nu_{trans,b}}{1+e_{trans}\cos\nu_{trans,b}}, \qquad (refer\ \ eq.(2.93)\ and\ \ (2.94))$$



$$\Rightarrow \Delta v_b = \sqrt{v_{trans,b}^2 + v_{final}^2 - 2v_{trans,b}v_{final}\cos\varphi_{trans,b}}$$

$$\Rightarrow \Delta v = |\Delta v_a| + |\Delta v_b|$$

$$\cos E = \frac{e_{trans} + \cos v_{trans,b}}{1 + e_{trans} \cos v_{trans,b}}, \qquad (refer\ \ eq.(2.9),\ E\ is\ \ eccentric\ \ anomaly\ )$$

$$M = n(t-T) = E - e\sin E, M : mean\, anomaly,\ n : mean\ \ motion, T : time\, of\ \ periapsis\ passage$$

$$M_0 = n(t_0 - T) = E_0 - e\sin E_0,$$

$$M - M_0 = n(t - t_0) = \left(E - e\sin E\right) - (E_0 - e\sin E_0)$$

$$\tau \equiv (t - t_0) = \frac{1}{n}\left[\left(E - e\sin E\right) - (E_0 - e\sin E_0)\right]$$

$$where\ \ n = \sqrt{\mu / a_{trans}^3},\ \ E_0 = 180^o\ at\ apoapsis,\ \ E_0 = 0\ at\ periapsis$$

$$\Rightarrow \tau_{trans} \equiv \sqrt{\frac{a_{trans}^3}{\mu}}\left[2k\pi + E - e_{trans}\sin E - (E_0 - e_{trans}\sin E_0)\right]\ ,$$

$$where\ \ E_0 = 180^o\ at\ apoapsis,\ \ E_0 = 0\ at\ periapsis$$

TABLE 6-1. Comparison of Coplanar Orbital Transfers. This table presents results of the trade-off between change in velocity and time of flight for different types of orbital changes. Recognize that the initial and final orbits are circular, whereas the transfer orbits are elliptical, with perigee and apogee values as shown. Notice that it's almost as hard to go to geosynchronous as it is to the Moon. You can verify this from Fig. 6-6.

	Initial Alt (km)	Final Alt (km)	v_{trans_b}	Bi-elliptic Transfer Alt (km)	Δv (km/s)	τ_{trans} (h)
		Transfer to Geosynchronous				
Hohmann	191.344 11	35,781.35			3.935	5.256
One-tangent	191.344 11	35,781.35	160°		4.699	3.457
Bi-elliptic	191.344 11	35,781.35		47,836.00	4.076	21.944
		Transfer to the Moon				
Hohmann	191.344 11	376,310.00			3.966	118.683
One-tangent	191.344 11	376,310.00	175°		4.099	83.061
Bi-elliptic	191.344 11	376,310.00		503,873.00	3.904	593.919

6.3.4 General Transfers

- The most general transfer is one using two non-tangential burns
- This transfer employs any orbit type because there are no restrictions for flight path angle or tangency requirements
- In practice, must solve the Lambert's problem (Section 7.6)

6.4 Non-Coplanar Transfers

- It may be necessary to change the inclination (i) and the longitude of ascending node(Ω) of an orbit, which requires applying Δv out of the orbital plane (non-tangential change)
  - Intercept or rendezvous maneuvers
  - Orbit orientation adjustment after launch
  - For a direct launch, the launch site latitude must be less than or equal to the desired inclination

6.4.1 Introduction

We need noncoplanar manueuvers mainly because

(1) A launch-site location restricts the initial orbit inclination for the satellite

(2) The direction of the launch can influence the amount of velocity the launch rocket must supply, so certain orientations may be more desirable

(3) Timing constraints may dictate a "launch window" that is not the best, from which we make changes.

Geometry for Launching a Satellite

A "direct launch" is a launch from the ground directly into the desired orbit.

For a direct launch, the launch-site latitude must be less than or equal to the desired inclination; otherwise we must change the inclination in orbit.

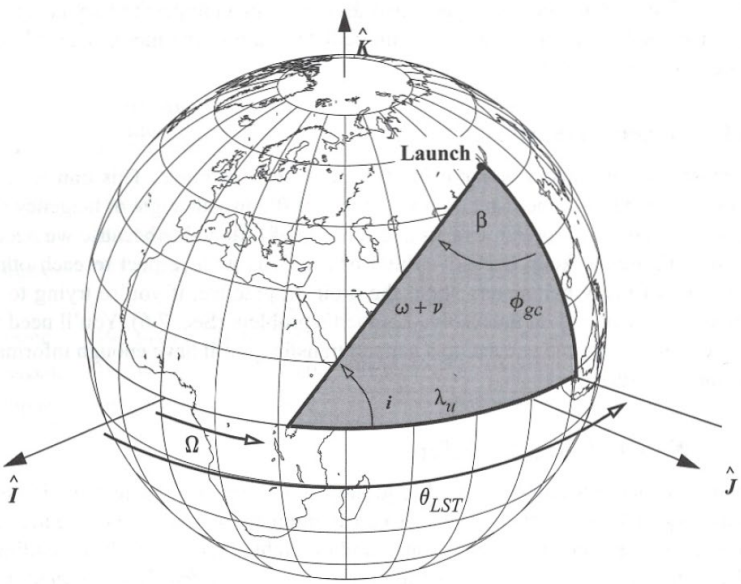


Figure 6-8. **Launch-Window Geometry.** Direct launches can occur only if the launch site's latitude is less than or equal to the satellite's inclination, i. Determining the launch window involves three basic steps: (1) finding the launch azimuth, β, from the site (I've shown an equivalent angle in the triangle because β is measured from north), (2) determining the auxiliary angle, λ_u, and (3) finding the GMST (or LST).

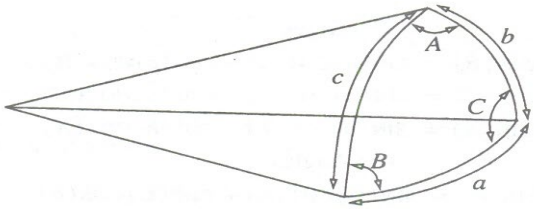


Figure C-4. Oblique Spherical Triangle. Oblique spherical triangles have no 90° angles. Also, angles and distance are interchangeable when dealing with the triangle.

$$cosB=-cosCcosA+sinCsinAcosb$$

From the spherical triangle laws, we obtain

$$cos(i)=cos(\phi_{gc})sin(\beta), \qquad sin(\beta)=\frac{cos(i)}{cos(\phi_{gc})}$$

Because magnitude of $sin(\beta) \leq 1$, the launch latitude (ϕ_{gc}) must less than or equal to the desired inclination. If we don't meet this requirement, we cannot launch the satellite directly into the desired orbit, so we may need to change the inclination.

Figure 6-9 shows several launch sites.

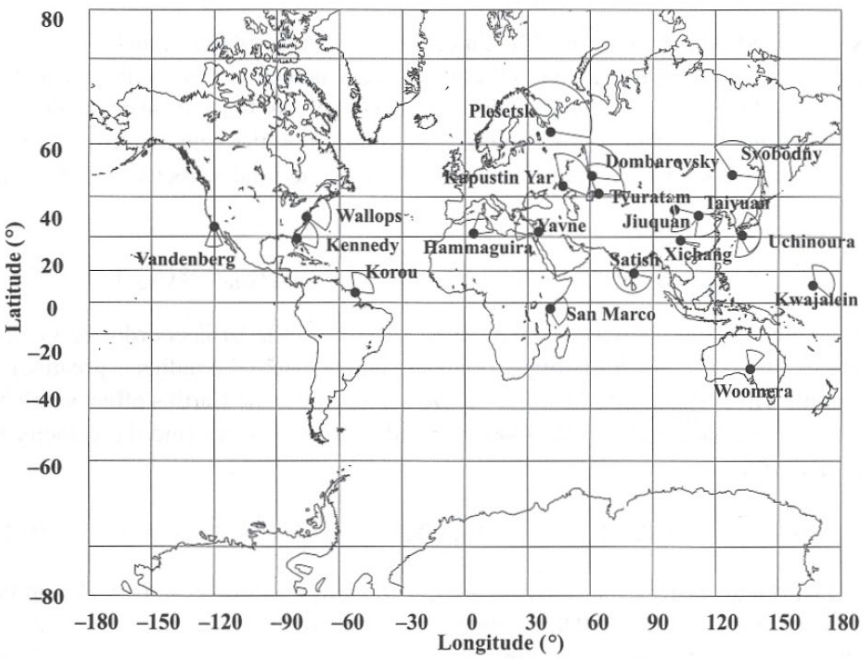


Figure 6-9. Locations for Selected Launch Sites. Direct launches are shown from various launch sites. Note the restrictions so boosters don't hit civilian areas and the relatively uniform low latitudes to take advantage of the Earth's rotation. Nominal 1000 km "fans" are shown for each site.

6.4.2 Inclination-only Changes

- To change orbit inclination only, we must change it at one of the two nodes (an equator crossing), which are the only two common points in the old and new orbits
- The size and shape of the final orbit are identical to those of the initial orbit: $v_{initial} = v_{final}$

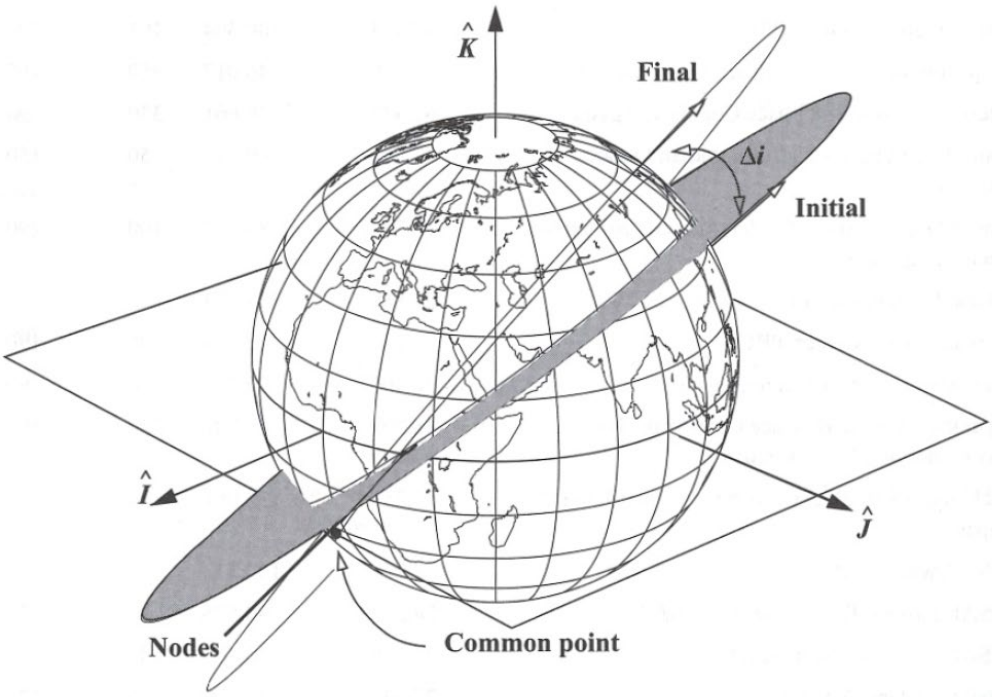
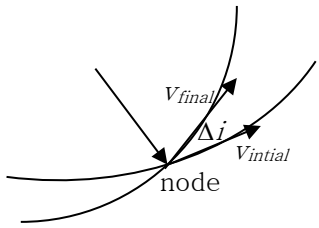


Figure 6-10. Inclination-Only Changes. An inclination-only change must occur at one of the nodal crossings because these are the only two points common to *both* orbits. The angular separation is equal to the change in inclination.

- Circular Orbit
  - Node direction is normal to the directions of $v_{initial}$ and v_{final} ⇒ changing velocity direction in the plane normal to the node vector.



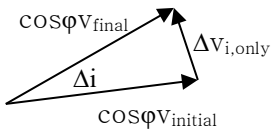
$$sin\left(\frac{\Delta i}{2}\right)=\frac{\Delta v_{i,only}/2}{v_{initial}}$$

$$\Delta v_{i,only}=2v_{initial}sin\left(\frac{\Delta i}{2}\right)$$

- Plane changes are expensive: $\Delta i=60^o \Rightarrow \Delta v_{i,only}=v_{initial}$ for circular orbits

- Elliptic Orbit

Cf. $tan(\varphi_{fpa})=\dfrac{esin(v)}{1+ecos(v)}$ (eqs. 2.93, 2.94)



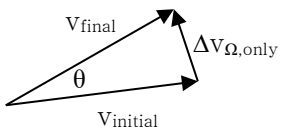
$$cos(\varphi_{fpa})v_{initial}=cos(\varphi_{fpa})v_{final}$$

$$\Rightarrow \Delta v_{i,only}=2cos(\varphi_{fpa})v_{initial}sin\left(\frac{\Delta i}{2}\right)$$

- For elliptic orbit, it is necessary to check both nodes to find smaller $v_{initial}$ because smaller $v_{initial}$ gives less $\Delta v_{i,only}$.

6.4.3 Changes in the Longitude of Ascending Node

- Only Circular Orbits are considered ⇒ One Burn.
  - Two possible common points to change longitude of ascending node only.



$$v_{initial}=v_{final} \quad , \quad i_{initial}=i_{final}=i$$

$$\Delta v_{\Omega,only}=2v_{initial}sin\left(\frac{\theta}{2}\right) \Rightarrow \theta=?$$

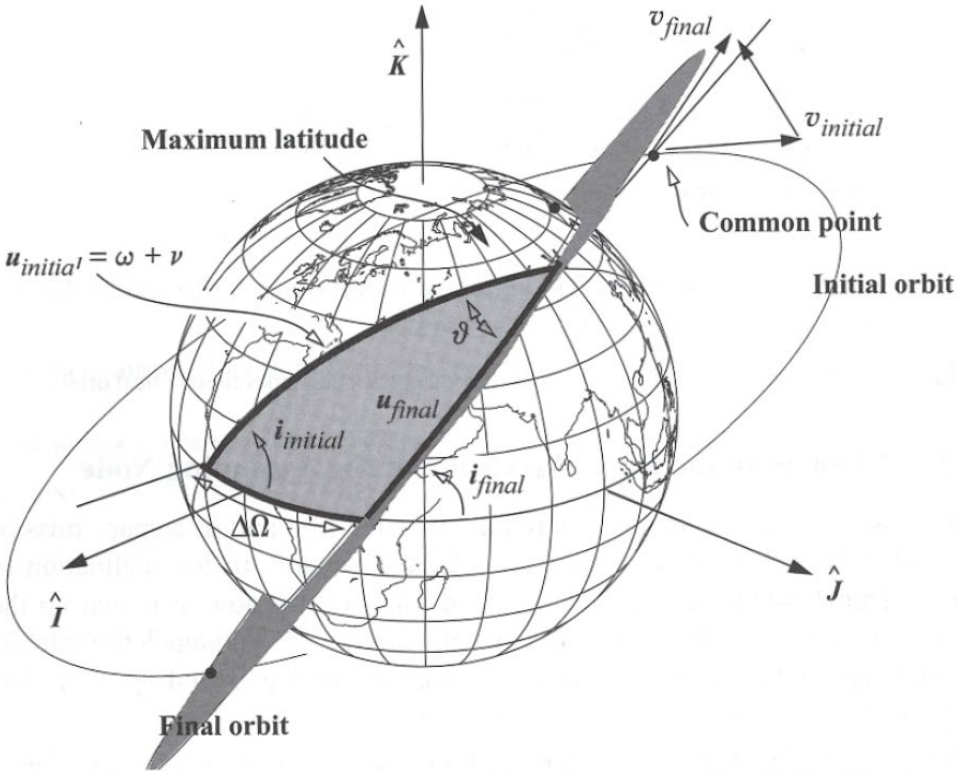


Figure 6-11. Effect of Changing the Right ascension of the Ascending Node. Circular orbits will always have two common points where the orbit paths cross each other. Because these points occur at different true-anomaly values in an elliptical orbit, we must analyze them separately.

- Use of Spherical Trigonometry

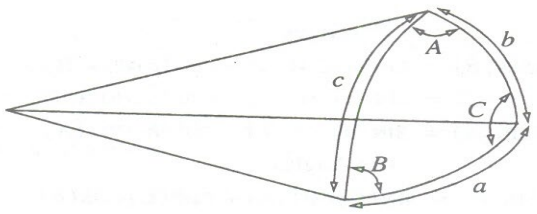
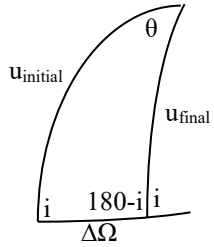


Figure C-4. Oblique Spherical Triangle. Oblique spherical triangles have no 90° angles. Also, angles and distance are interchangeable when dealing with the triangle.

$$\cos A = -\cos B \cos C + \sin B \sin C \cos a$$
$$\sin A \cos b = \cos B \sin C + \sin B \cos C \cos a$$
$$\sin C \cos a = \cos A \sin B + \sin A \cos B \cos c$$



$$\cos\theta = -\cos i\cos(180-i) + \sin i\sin(180-i)\cos(\Delta\Omega)$$
$$= \cos^2 i + \sin^2 i\cos(\Delta\Omega) \Rightarrow \theta$$

$$\sin\theta\cos(u_{final}) = \cos i\sin(180-i) + \sin i\cos(180-i)\cos(\Delta\Omega)$$
$$\cos(u_{final}) = \frac{\cos i\sin i(1-\cos(\Delta\Omega))}{\sin\theta} \Rightarrow u_{final}$$

$$\sin(180-i)\cos(\Delta\Omega) = \cos\theta\sin i + \sin\theta\cos i\cos(u_{initial})$$
$$\cos(u_{initial}) = \frac{\sin i\cos(\Delta\Omega) - \cos\theta\sin i}{\sin\theta\cos i} = \tan i\,\frac{\cos(\Delta\Omega) - \cos\theta}{\sin\theta} \Rightarrow u_{initial}$$

- Using $u_{initial}$ and u_{final} above enables us to find the common point

- As elliptic orbits have complicated maneuvers (generally multiple burns), it is much more appropriate to solve the Lambert's problem.

6.4.4 Changes to Inclination and Longitude of Ascending Node

- Only Circular Orbits are considered ⇒ one burn
  - This change may appear to be similar to the preceding case, but the firing position in the orbit differs

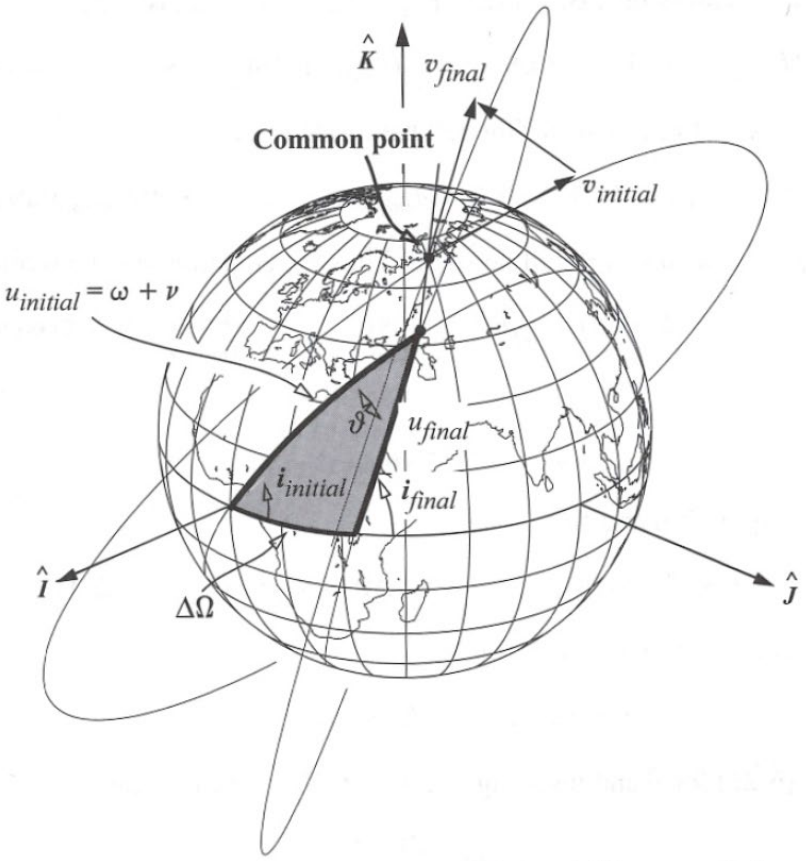
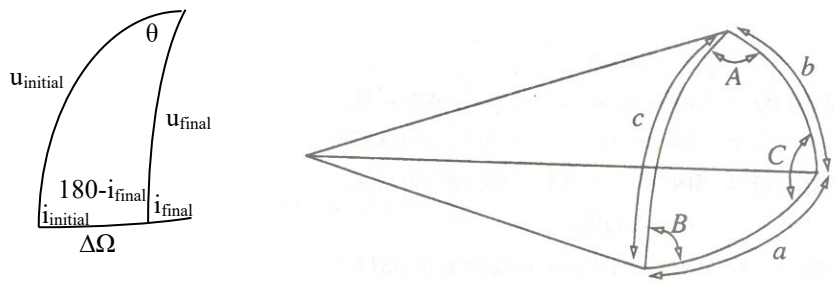


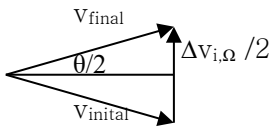
Figure 6-12. Geometry for General Noncoplanar Change. This type of change requires a general solution to find the common point of the two orbits. Note that both the inclination and ascending node are changing.



$$\cos A = -\cos B \cos C + \sin B \sin C \cos a$$

$$\cos\theta = -\cos i_{initial} \cos(180 - i_{final}) + \sin i_{initial} \sin(180 - i_{final}) \cos(\Delta\Omega)$$

$$= \cos i_{initial} \cos i_{final} + \sin i_{initial} \sin i_{final} \cos(\Delta\Omega)$$



$$\Delta v_{i,\Omega}=2v_{initial}\sin(\frac{\theta}{2})$$

$$\sin A\cos b=\cos B\sin C+\sin B\cos C\cos a$$

$$\sin\theta\cos(u_{final})=\cos i_{initial}\sin(180-i_{final})+\sin i_{initial}\cos(180-i_{final})\cos(\Delta\Omega)$$

$$\cos(u_{final})=\frac{\cos i_{initial}\sin i_{final}-\sin i_{initial}\cos i_{final}\cos(\Delta\Omega))}{\sin\theta}\Rightarrow u_{final}$$

Similarly,

$$\sin C\cos a=\cos A\sin B+\sin A\cos B\cos c$$

$$\cos(u_{initial})=\frac{\sin i_{final}\cos(\Delta\Omega)-\cos\theta\sin i_{initial}}{\sin\theta\cos i_{initial}}\Rightarrow u_{initial}$$

- Using $u_{initial}$ and u_{final} enables one to find the common point